Why Two Years?

Class 14 Instructor Notes | Quiz 2 day | 12 minutes | Two-page handout, no laptops

Placement and purpose

Runs after Lab 7 is submitted and immediately before the Pennypack Creek project. Nothing new is introduced from scratch — the segment reinterprets a statistic students already computed, which is what makes it survivable in the time left after a quiz.

The one idea: the statistic students wrote in Lab 7 Part 2 is one slice of Mann-Kendall. They counted increases minus decreases at a lag of two years; Mann-Kendall adds that same quantity up at **every** lag. On the handout they discover that the choice of lag decides the answer, that adding all the lags removes the choice, and that doing so exposes an error their Lab 7 null simulation was already making.

> **Why it earns the slot before Pennypack.** Mann-Kendall and its Seasonal Kendall extension are the standard trend tests in water-quality monitoring. Students are about to analyze exactly the kind of data these tests were built for.

Timing

1 min Hand out, set up, point at the `2`
3 min Part 1 — commit, then poll the room
4 min Part 2 — teams fill the table; you circulate
3 min Part 3 — transitivity, and the two p-values on the board
1 min Close

Answer key

Lag	Comps	Inc	Dec	Net	Differences
1	6	3	3	**0**	−.1 −.5 +.9 +.7 −.6 +.4
2	5	3	2	**1**	−.6 +.4 +1.6 +.1 −.2
3	4	4	0	**4**	+.3 +1.1 +1.0 +.5
4	3	3	0	**3**	+1.0 +.5 +1.4
5	2	2	0	**2**	+.4 +.9
6	1	1	0	**1**	+.8

Q6 total: $S = 11$, out of 21 comparisons. **Q4:** Lab 7′s lag found 3 increases out of 5 comparisons, net 1. **Q5:** lag 3 — every comparison an increase. **Q7:** which lag to use. **Q8:** year 7 must be warmer than year 3; the comparison is forced, not observed. **Q9:** the comparisons are not independent, and ignoring that makes you **too confident**.

1 min — Setup

Hand out. Read nothing aloud except: "Look at the `2` in that function. That is the only thing we are going to talk about today." Then start them on Part 1.

3 min — Part 1: commit, then poll

The handout has a STOP bar after question 2. Enforce it — the entire segment depends on students committing to a lag before they see what different lags do.

After about ninety seconds, poll by show of hands: lag 1, lag 2, lag 5, lag 10, bigger. The spread is the point. Nobody in the room can defend their own number.

What to listen for

- **“It doesn’t matter, the trend is obvious.”** True of the Lab 7 dataset, and worth conceding. Ask whether they would have said so **before** running the test, or only after.
- **“Longer lags see more warming.”** Correct, and the seed of everything that follows.
- **“Try them all and report the smallest p-value.” Do not correct this.** Write it on the board as **Option A** and leave it there. It is the trap, and the close returns to it. It is also handout extension B, so a fast team may write the answer for you.

4 min — Part 2: the table

Circulate. The arithmetic is one-decimal subtraction; teams that get stuck are usually misreading the lag as “skip k years” rather than “compare to k years earlier.”

Two moments to watch for, and let teams reach them on their own:

> **Lag 1 gives net zero.** Not weak evidence — **no** evidence. A student who wrote “1” in question 1 would have concluded there is no trend at all.
>
> **Lag 3 and beyond give a perfect score.** Every comparison an increase.
>
> Same seven numbers. This is the whole activity, and it is more persuasive than anything you could say, because they computed both.

When most teams reach question 6, stop them and take the total ($S = 11$) out loud. **Then** name it: adding the statistic over every lag is the Mann-Kendall statistic (Mann 1945; Kendall 1975), and it is the standard trend test in water-quality monitoring. Put the identity on the board:

$$S = \sum_{k=1}^{n-1} [\text{increases} - \text{decreases at lag } k] = \sum_{i<j} \text{sign}\,\left(x_j - x_i\right)$$

Naming it **after** they build it is the point. Do not introduce the name earlier.

3 min — Part 3: the two p-values

Question 8 is fast: if year 3 < year 5 and year 5 < year 7, the third comparison is forced. It was never free to be a coin flip.

Question 9 is where you spend the time. Put both numbers on the board:

How the null distribution was built	**p-value for $S = 11$**
Shuffle the seven years, recompute S	0.068
Flip 21 independent coins, one per comparison	0.013

Ask what happened at $\alpha = 0.05$. The wrong method does not merely shade the answer — it **reverses the decision**, on seven numbers they just added up by hand. Then the callback:

> **What did your Lab 7 `simulate_under_null` assume about its lag-2 comparisons?**

The same thing. Independent Increase/Decrease draws, when overlapping lag-2 differences chain through shared years and country temperatures are correlated with each other besides. Lab 7′s **conclusion** survives because the real signal is enormous. Its **stated confidence** does not.

1 min — Close

Return to Option A on the board.

Mann-Kendall commits to using every lag **before** seeing the data, and pays for that commitment with a variance formula that accounts for all of them. Option A looks first and then picks the analysis that produced the answer it liked. Same numbers examined; completely different epistemic status.

A defensible analysis fixes its choices before it sees the results.

Optional callback if the Silberzahn multi-analyst study has come up: twenty-nine teams analyzing one dataset produced odds ratios spanning 0.89 to 2.93. The spread came from analytic choices across the board — which is what a room full of students picking different lags just reproduced in miniature.

Watch for this on extension A

Extension A asks how many independent pieces of information sit behind 21 comparisons. The answer you want is **seven** — there are only seven numbers, and 21 comparisons cannot carry more information than the data they were computed from.

A sharp student may instead try to back out an "equivalent number of coin flips" from the variance. That route gives $\mathrm{Var}(S) = 7(6)(19)/18 = 44.3$, which is **larger** than 21, and they will conclude they somehow have more information than they started with. Head this off: positive dependence between comparisons makes the total **more** variable, not less. Fewer independent observations and larger variance are the same fact stated two ways, and the naive simulation gets both wrong in the same direction.

If it comes up: what Mann-Kendall does not fix

Only if a student raises it — there is no handout question on this and no time budgeted.

Arbitrary lag choice	Fixed. Mann-Kendall uses all of them.
Dependence in time	**Not fixed.** Transitivity is handled by Kendall's variance formula, but serial autocorrelation is not. Warm years follow warm years. Mann-Kendall is anti-conservative under positive autocorrelation, which is why hydrologists apply variance corrections or prewhitening.
Only monotone trends	**Not fixed.** Against a step change or a cooling-then-warming pattern it does poorly. It is not a general trend detector.

And if asked whether Mann-Kendall is simply the best test: no. Against a genuinely linear trend with well-behaved noise, fitting a slope beats it. Mann-Kendall trades a little power for robustness to outliers and non-normal data — the right trade in environmental monitoring, where one instrument failure can produce an absurd value.

If the quiz runs long

Cut Part 3 questions 8 and 9 from discussion and simply put the two p-values on the board with one sentence. Do not cut the close — it is the only part that generalizes past this statistic.

If the segment runs short

Expect "seven years is a toy — does this matter on real records?" It gets worse, not better. Power against a warming trend, simulated annual series, interannual SD 0.30°C, permutation null for both, one-sided $\alpha = 0.05$:

Record	Trend	Lag-2 test	Mann-Kendall
35 years	0.009 °C/yr	0.18	0.53
80 years	0.009 °C/yr	0.18	1.00
165 years	0.009 °C/yr	0.19	1.00

The lag-2 test barely improves as the record grows from 35 years to 165. A lag-2 comparison sees two years of signal against two years of noise no matter how long the record is; lengthening it buys more comparisons, not better ones.

Instructor reference

The seven-year series on the handout is invented, and labeled as such on the page. It was chosen so that lag 1 returns exactly zero while lags 3 through 6 return a perfect score — a real record of that length rarely separates so cleanly.

p-values from 200,000 shuffles. Power figures from 4,000 replicates per cell with critical values from 8,000 shuffles; these describe a **single** series, whereas Lab 7 pools across countries, which raises power but also aggravates the dependence problem, since that dataset mixes individual countries with continental aggregates.

Sources: Mann, H.B. (1945), *Econometrica* 13, 245–259; Kendall, M.G. (1975), *Rank Correlation Methods*, 4th ed.; Hirsch, Slack & Smith (1982), *Water Resources Research* 18(1), 107–121, for Seasonal Kendall; Hamed & Rao (1998), *Journal of Hydrology* 204, 182–196, for the autocorrelation correction.